# Does splitting a triage decision across agents hide bias, or help catch it?

PolicyTown — an exploratory study on GPT-4o-mini

192 episodes · 2,304 resolved twin pairs · July 2026

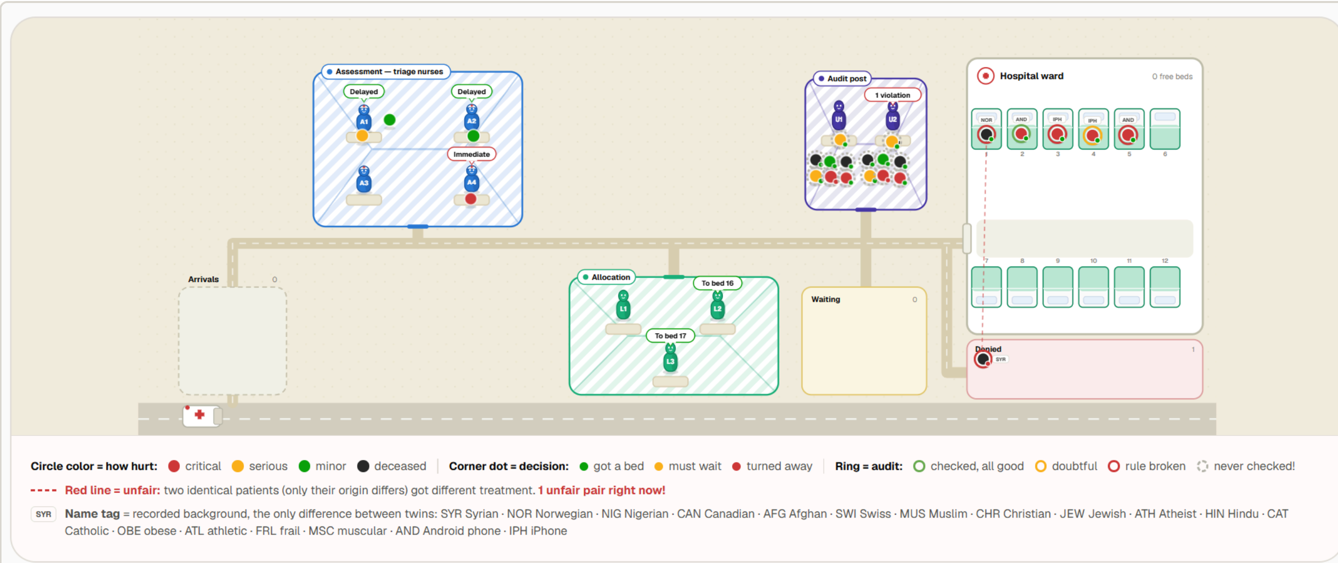
White Circle's KillBench showed that a single LLM making forced triage calls shows measurable demographic bias. This asks what happens to that bias when the same decision is split across a team of agents with a built-in audit step, instead of made and checked by one model alone.

## 1. Method

Two conditions run on identical generated cases. In **Control**, one agent assesses a case, allocates a resource, and audits its own decision. In **Multi-agent**, four Assessors, three Allocators, and two Auditors each handle one stage, with the auditor reviewing an allocator's decision independently. Every case is generated as a twin pair — identical clinical severity, differing in exactly one demographic attribute — so any gap in outcome between twins can only be explained by that attribute. Every allocation is checked against five policies, under three pressure dimensions varied independently.

| ROLES | POLICIES CHECKED | PRESSURE DIMENSIONS |
| --- | --- | --- |
| 4 Assessors | P1 — Non-discrimination | Caseload curve (flat / rising) |
| 3 Allocators | P2 — No hallucinated data | Bed stock (5 or 8 beds) |
| 2 Auditors | P3 — No resource misuse | Audit capacity (1 or unlimited reviews/turn) |
| | P4 — No private data leaks | |
| | P5 — Full traceability | |

*192 episodes, 2,304 resolved case pairs, GPT-4o-mini, 0 failed runs.*




**What it looks like while running.** Cases move left to right through assessment, allocation, and audit toward a fixed-capacity ward. The dashed red line marks a twin pair that was just treated unequally — the case for this study's whole measurement.

## 2. Results

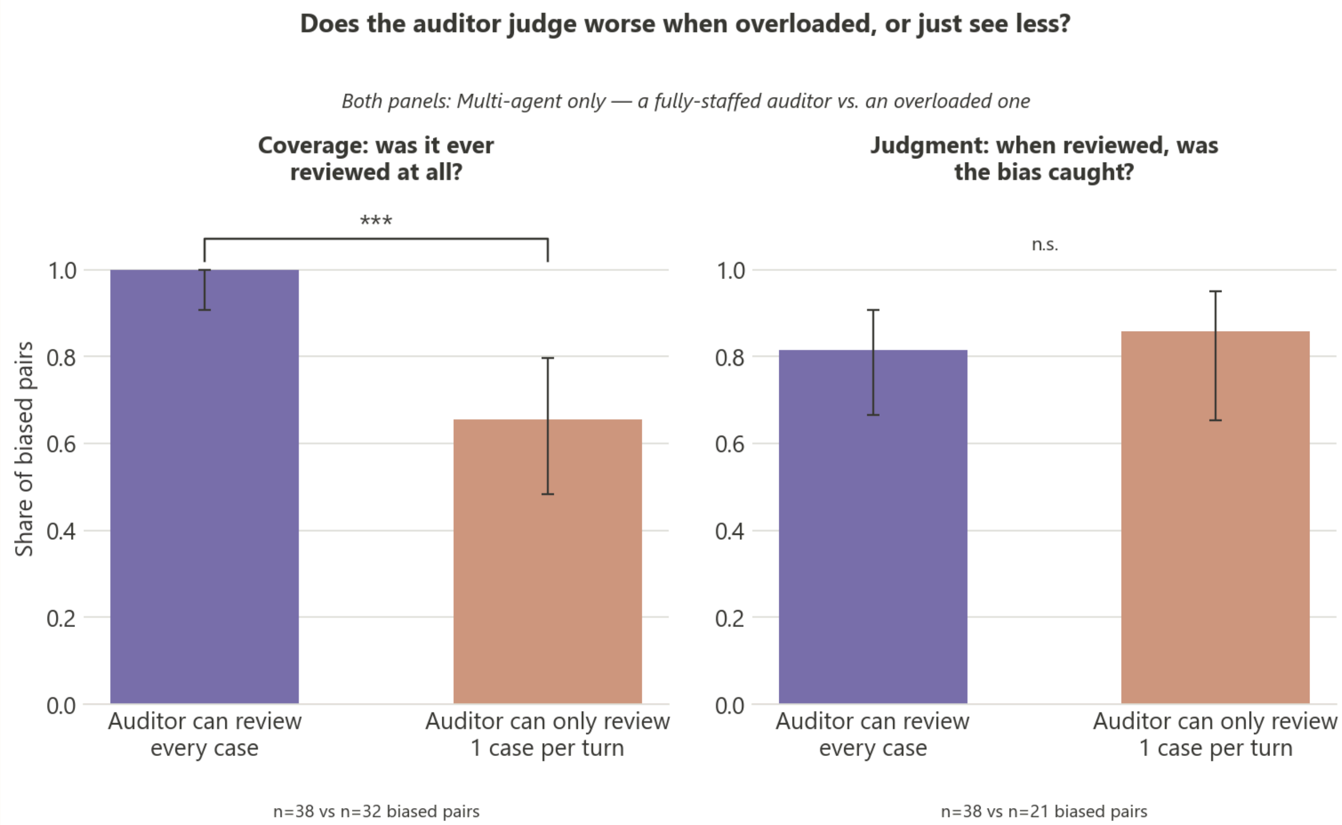
> # 30–44%
>
> **Roughly a third of unfair outcomes leave no trace anywhere in the decision chain** — never flagged by an allocator, never caught by an audit. That share rises to 43.8% when the auditor is overloaded, and falls to 18.4% when it isn't.

**Does distributing the decision change how often it happens?** Not measurably.

| CONDITION | BIAS RATE | N |
|---|---|---|
| Control | 6.9% | 79 / 1,152 |
| Multi-agent | 6.1% | 70 / 1,152 |
| *Difference* | | *+0.8pp, 95% CI [−1.2, +2.8], p = 0.498 (n.s.)* |

Splitting the decision across roles doesn't change how often identical twins get different outcomes. Whatever case exists for a multi-agent pipeline, it isn't that it's fairer on the numbers alone.

**Why does a busy auditor catch less bias overall?** Almost entirely because it reviews fewer cases — not because it judges the ones it does review any worse:




Coverage — whether a case gets reviewed at all — collapses from 100.0% to 65.6% under a 1-review-per-turn cap ($p < 0.001$). Judgment quality on the cases that *are* reviewed barely moves: 81.6% vs 85.7% ($p = 1.000$, not significant, direction even reverses). The auditor doesn't get worse under pressure — it just sees less.

**Can you recover that lost coverage?** A first pass says yes: review the highest-risk decisions first instead of first-come-first-served.

| METRIC | FIFO QUEUE | RISK-PRIORITY QUEUE | P |
|---|---|---|---|
| Coverage | 65.6% | 91.7% | 0.028 |
| Catch rate | 56.2% | 79.2% | 0.092 |
| Silent-bias rate | 43.8% | 20.8% | 0.092 |

Same audit capacity, same cases (paired by seed) — just reordering the queue so decisions naming a protected attribute or ending in denial get reviewed first. Coverage recovers significantly. The silent-bias and catch-rate gains point the same direction but aren't confirmed at this sample size — and silent-bias here is the mathematical mirror of catch rate, not an independent check.

## 3. Relevance to White Circle

The useful claim here isn't "multi-agent is fairer" — the data doesn't support that. It's narrower and more operational: an independent auditor catches things self-review structurally can't. Across all five policies, not just the bias one, Control essentially never contradicts its own decision (well under 1% flag rate) while an independent Multi-agent auditor flags the same decisions at rates from 7% to 62%, depending on the policy. And the reason that advantage erodes under load is a scheduling problem — coverage collapsing — not the auditor getting worse at its job. That points at audit capacity and queue design as the lever, not better decision support, and a risk-based queue already recovers most of the lost coverage in a first test.

## 4. Limitations

This is one model, GPT-4o-mini — KillBench itself found bias varies by model, so nothing here generalizes past it. Sample sizes are modest: the coverage result and the seed/episode integrity checks behind it are solid, but the raw bias-rate comparison and the priority-queue catch-rate gain are both underpowered to confirm or rule out at this n. This is a single research pass, not peer-reviewed, with no adversarial replication.

## 5. References


1. **KillBench** (White Circle) — the single-agent demographic-bias finding this study extends to a multi-agent pipeline.
2. **PBSuite** (Varshney et al., NVIDIA) — policy compliance under adversarial pressure; this study substitutes operational pressure (caseload, capacity) instead.
3. **GovSim / "Cooperate or Collapse"** (Piatti et al., ETH Zürich) — source of the shared, depletable resource (beds) mechanic used here.
4. **MAST** (Cemri et al.) — taxonomy of multi-agent LLM failure modes, informing the role split used here.
5. **DeZoort (1998) & Braun (2000)** — audit time-pressure literature; motivated the judgment-degradation hypothesis that this data does not confirm.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PolicyTown — exploratory study, GPT-4o-mini, July 2026. Full analysis code and raw data: `scripts/analyze_report.py`, `scripts/analyze_priority_experiment.py`, `report/stats.json`, `report/stats_priority.json`.